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Forward Deployed Engineer · AI Solution Architect

AI and data systems for regulated industries, built with public and private investors as the users

Caltech AI/ML Engineering Certification ◆ Stanford Master's ◆ Tufts BA

Forward deployed by track record: I lead the discovery workshop and the architecture review, deploy hands-on, write the implementation guide, and stay through production. A decade on enterprise data platforms. At Nielsen/Gracenote I owned the architecture for the unified platform behind a $300M-a-year data product: ~30 billion records, ~200 million daily updates on Kafka, and entity resolution across conflicting source catalogs. At HP I led teams of up to 15 on the API data exchange and governance platform for a $40B partner program. Today I build AI on top of the data and I validate what it says: a production investor platform and private CRM with a voice interface and a governed AI assistant, built inside an agent harness where models from other families judge every risky change; and high-ROI risk-prevention decision support for defense sites (extreme weather), whose data warehouse escrows every source and enforces provenance in the type. Thirteen years in tech: I ran data-platform programs alongside engineering teams at HP and Nielsen, moved to hands-on engineering in 2023, and have built three systems solo since, one in production. I use AI coding agents daily under enforced controls.

PROFESSIONAL EXPERIENCE

Defense Investors Network (DIN) and Defense Angel Network (DAN) Palo Alto, CA | May 2026 – Present

Forward Deployed Engineer / AI Solution Architect — via Ripley Decision Advantage · embedded with the team

- **LYNDA, production investor-relations platform.** Sole engineer on a private CRM and AI assistant for a dual-use defense-investor team (maritime industrial base, critical-mineral sovereignty, energy resilience), live with a real user base on web and iOS (TestFlight) with voice and chat interfaces. Human-in-the-loop by design: no record is changed, sent or deleted without a team member's confirmation, and every action is written to an audit trail. Requirements gathered directly from the operating team.
- **Data architecture.** Selected PostgreSQL with recursive CTEs for relationship traversal after evaluating Spark as over-scaled for the data volume; added Neo4j/Cypher where the relationship graph justified it, implementing deterministic, auditable path-scoring with hard safety gates for the warm-introduction engine.
- **Access control and PII protection (NIST AI RMF, SR 11-7).** Visible / redacted / hidden tiers enforced at the query layer; a reveal-field whitelist makes name, email, phone and notes unleakable on a redacted row; PII masked in storage, not only at the model boundary; a missing permission returns nothing, so a record's existence cannot be inferred.
- **Audit trail and model evaluation.** A SHA-256 hashed-payload audit trail (canonical payload never stored in plaintext) with per-field old/new diffs through a changelog endpoint; attack drills on every release in which a model from a different company tries to trick Lynda and grades how it held; an after-answer truth check that compares what actually changed in the CRM with what the model said happened and flags any claim it cannot back up.
- **AI assistant architecture.** 36 registered tools behind Pydantic-validated dispatch, a bounded execution loop and a content-policy scrubber, on a vendor-neutral design so domain policy ports across model hosts and A2A/MCP topologies; the confirm step runs from the person's tap, never through the model.
- **Release gating and code review (agent harness).** Every change ships as a pull request I merge, authored by a coding agent under its own GitHub identity; three models from outside the authoring model's family review every risky plan and diff before merge; research agents return data files whose verdict rows pass a validator before human review; 1,700-plus backend tests pin each invariant before the fix that closes it.
- **Compliance mapping and ontologies.** Every control mapped to NIST CSF 2.0 (the one open function, Respond, named on the roadmap rather than hidden) and NIST AI RMF. Contacts, funds and deals carry FIBO; signals carry DoD program-element and Product Service Code identifiers, and every demand signal sits on the federal funding ladder.

Stack: FastAPI · PostgreSQL (recursive CTEs) · Neo4j/Cypher · Claude Code · GCP (Confidential Space · Cloud KMS · Cloud Storage · Cloud Run · Cloud SQL) · Google OAuth 2.0 · AES-256-GCM · Fernet · Ed25519 · cosign · pytest · promptfoo · OpenTelemetry · ElevenLabs/Chirp · iOS (TestFlight)

EGON — Ripley Decision Advantage Palo Alto, CA | June 2026 – Present

Founder & Architect — extreme-weather risk decision support for infrastructure owners and investors · built and piloted, four countries

- **Decision-support system.** Architected EGON, a service-life extender for public infrastructure and an exit-value enhancer for investors: prices what extreme weather will cost a specific asset, what protects it, and what that protection is worth, and returns exportable scenarios for the owner's own models. Export-only integration; no connection to financial cores. Piloted across ports, data centers and district infrastructure in four countries.
- **Escrowed data warehouse.** Separated source curation (admit, fetch, escrow, refresh) from answer assembly, with the assembler read-only. Escrowed 45 datasets (39 checksummed) on 90-day to 5-year cadences, each fetched once, hashed, vintage-pinned and monitored for mutation, across NOAA Storm Events / SWDI, OpenFEMA, ERDDAP, FRED, BLS, Census and USDA NASS; upstream changes surface as findings rather than silent diffs.
- **Typed provenance.** Enforced citation at the type level so no figure can be constructed without its source; every number in an output traces to a challengeable record.

- **Classification-boundary egress control.** Implemented a per-request data-class guard evaluated on every attempt, including retries and fallbacks; identified and closed a default-permit path in the bundle-cutting tool (glob selection in place of an allowlist), replacing it with complete mediation and default deny.
- **Data tiering by custody.** Classified data as never-held, entered-but-never-ingested, opt-in, derived-and-exportable, or engagement-fact, removing entire regulatory gates by not ingesting records the system does not need.
- **Multi-agent orchestration.** Built on Google ADK with a full A2A implementation, including custom message-envelope, validation and resilience layers (circuit breakers, exponential-backoff retries, session caching). Worker agents: extreme-weather risk analyzer, nature-based-solutions (45+ strategies), cost/benefit (statistically significant improvements only), and verification with geocoding validation.
- **Delivery.** Export-only FastAPI REST endpoints and a Vanilla JS / Chart.js / WebSocket dashboard; every figure traceable to its escrowed source.

Stack: Google ADK · Google A2A protocol (custom envelope, validation & resilience layer) · escrowed data warehouse (45 datasets, 39 checksummed; NOAA, OpenFEMA, ERDDAP, FRED, BLS, Census, USDA) · typed provenance · classification-boundary egress guard · FastAPI (export-only REST) · Vanilla JS / Chart.js / WebSockets · GCP · Python

X, the moonshot factory (Alphabet) Palo Alto, CA | 2025 – May 2026
AI Engineer, stealth project — via Ripley Decision Advantage

- Agentic data management across 100+ data sources inside an AI lab. Google ADK, A2A protocol (custom envelope, validation & resilience layer), FastAPI (export-only REST), GCP, Python.

SpinF AI San Francisco, CA | Fall 2024
Sales & ML Engineer, Insurance Vertical — via Ripley Decision Advantage · seed-stage startup

- **Document pipeline.** A high-precision document-extraction and comparison pipeline over unstructured insurance contracts: policy interpretation, redlining, and detection of coverage gaps and high-risk wording.
- **Voice AI.** A real-time voice AI assist tool for live sales/CX calls (fine-tuned LLMs, ASR/TTS, agent-to-human handoff).
- **Discovery to deployment.** Opened the insurance vertical from zero: technical discovery, executive demos, and deployment to Tier-1 accounts.

Stack: Fine-tuned LLMs · ASR/TTS (ElevenLabs, Groq/Wispr, Chirp) · Document-extraction pipelines · Contract intelligence · NAIC alignment

Ares Management / BootstrapLabs VC San Francisco, CA | 2024
Forward Deployed Engineer & Technical/Product Lead — via Ripley Decision Advantage

- **Trust-scoring graph.** Engineering and product manager for a nodes-first knowledge-graph platform quantifying trust and credibility across enterprise data ecosystems; engineered entity-interaction models (nodes/edges) for real-time trust scoring of open-source and commercial data sources in Java and Node.js. Concluded at prototype stage; did not reach production.
- **Positioning.** Against incumbents (Collibra, Qlik, Monte Carlo, Credo AI) on behavioral trust metrics and verifiable data lineage; led investor pitches and prototype engineering for PE/VC use cases.

Stack: Knowledge graphs (nodes/edges) · Java · Node.js · Data lineage & provenance · RAG · SEC/FINRA compliance

Ripley Decision Advantage Palo Alto, CA | 2023 – 2024
Founder & Engineer — self-funded R&D

- **Compound AI System.** Solo-designed and built a working multi-agent data-management platform for an OSINT application, demoed live on an unclassified cloud instance, every cluster running, not just designed. Built while exploring a prospective application that did not materialize.
- **Two-layer orchestration.** LangChain as the central router (the “RipleyOrchestrator”), routing tasks across local 7B/13B models and passing context along the ingestion → RAG → Neo4j truth-anchor path; SuperAGI as the autonomy engine for worker nodes running continuous, multi-step loops. LangChain directed the traffic; SuperAGI drove the cars.
- **Ingestion cluster.** An Event Ingestion agent (continuous targeted scraping), a Data Broker (on-demand onboarding and synthetic data), and a Data Pruning agent that compressed and chunked intel to fit 2023’s small context windows.
- **Analytical agents.** A Red Team agent that generated predictive models of adversary behavior from OSINT (not just retrieval), a Blue Team agent that mapped structural policy dynamics, and a retrieval (RAG) agent for real-time classification on an Elasticsearch kNN vector index.
- **Graph-grounded verification.** Applied to model outputs rather than retrieval: generative outputs routed through a Neo4j graph to validate their logic and suppress any relationship not supported by known facts, killing hallucinations before they reached the user.
- **Multi-cloud burst compute.** Shadeform allocating across GCP, AWS and Azure on demand, in a two-stage pipeline: agents scraped and synthesized public USG policy and unclassified OSINT in the cloud; only final artifacts were packaged for the customer’s own on-prem / edge environment.
- **Privacy Guard and Documentation agents.** Automated redaction and encryption for safe data blending; a tool-executing agent that bound generative decisions to rigid macros to populate official templates, rather than free-form generation.

Stack: Compound AI System · Multi-agent orchestration (SuperAGI, LangChain) · Local open-source LLMs (LLaMa, MPT-7B) · Elasticsearch (kNN vector search / RAG retrieval) · Shadeform (multi-cloud: GCP / AWS / Azure) · Neo4j · graph-grounded output verification · Python

Nielsen / Gracenote Emeryville, CA | 2021 – 2023
Director, Data Strategy & Execution — data platform architecture on AWS

- **Restarted a stalled platform.** Restarted and shipped the unified data platform behind the world's largest video-metadata provider's $300M-a-year core data product after a five-year impasse, by identifying the new revenue products and upsell paths the platform could unlock and making the case that moved the organization to finish it. The blocker was never buildability; it was the connection to commercial value, which I supplied.
- **Data architecture and consolidation.** Owned the data architecture and schema/model decisions and coordinated the engineering build to delivery; re-architected the semantic knowledge infrastructure so heterogeneous source catalogs (conflicting entity IDs, schemas, formats) resolved into one canonical, auditable model, consistent and audit-grade across high-volume ML/AI workloads.
- **Real-time pipelines at scale.** ~30 billion records and roughly 200 million daily updates on Kafka: real-time event processing on the streaming path, batch for bulk application; Cassandra and AWS.
- **Technical leadership.** Founded Technical Women in Gracenote, doubling female-engineering retention in year one.

Stack: AWS · Kafka · Cassandra · BigQuery · Data architecture & schema design · Entity resolution & data fusion · Semantic knowledge infrastructure · Model Risk Management · Data provenance · ML/AI workload orchestration

HP Palo Alto, CA | 2018 – 2021
Senior Product Manager — Data Strategy / Data Transformation

- **Enterprise integration.** Shipped AMPLIFY Channel Partners Data Exchange, HP's first API-based data exchange and governance capability for its $40B channel-partner program, with data-contract models across telemetry sources and RBAC on downstream access.
- **Team leadership.** Led enterprise data-strategy programs targeting $15B in new revenue and cost savings; led teams of up to 15 across data architecture, API management and data-linking workstreams.

Stack: Enterprise data architecture · API management · Data linking & contracts · RBAC · Data governance · Python

Miller Strategic Design Palo Alto, CA | 2013 – 2018
Principal — Corporate Strategy & Product Design

- Contract engineering and product lead across a series of SV-startup engagements, including EdCast, CoLoadX (logistics SaaS) and SimpleFI (fintech). Full-stack engineering · MVP / rapid prototyping · SaaS architecture · Python.

LEADERSHIP & CONVENING

Rocky Mountain Chapter Lead (Volunteer), Defense Investors Network and Defense Angel Network · Supporter, Taiwan Chapter (launch 2026-11-07)

Chair, ONA Dialogues: three convenings, Bay Area (2026-09), Washington, D.C. (2026-10), New York City (2027-01)

Led the Wharton Aerospace Ventures panel "Crowding In Capital: Financing U.S. Reindustrialization" (Washington, D.C., 2026-10); leading the six-month LP/GP tabletop working group that follows

Advisor, AI Scenario-Based Tools — Office of Net Assessment, Department of War

Senior Fellow, Institute for State Effectiveness (Washington, D.C.) · Special Projects, Smith Richardson Foundation, International Security & Foreign Policy Program

Team Mentor, Hacking for Defense (H4D)

Founder, Rainbow Primary School (Hyderabad, India), 2005 – present, a community-led school still operating after 20+ years

EDUCATION & CERTIFICATIONS

AI/ML Engineering Certification — California Institute of Technology (Caltech), 2023-07 to 2024-04. Statistics, data science, machine learning, deep learning, Python; capstones built deep-learning classifiers and AI agents.

AI Solution Architect Master Course — Duc Haba via ELVTR, 2025-02 to 2025-04. AI project blueprints.

Master's, Design, Entrepreneurship & Education — Stanford University · **MA** — University of Connecticut · **BA** — Tufts University

TECHNICAL SKILLS INVENTORY

Data (Infrastructure & Analytics): Deployed: PostgreSQL (production core: array columns, recursive CTEs), Cassandra and AWS (Nielsen unified platform), Neo4j/Cypher (graph fact-checking, warm-introduction engine, EGON verification). Pipelines: escrowed sources (hash, vintage, mutation watch), typed provenance, ETL & ingestion (45 escrowed datasets; 20+ external-source loaders), entity resolution, data fusion, OSINT acquisition, lineage, data quality. Prior platform roles: BigQuery, Snowflake, Spark SQL, dbt, Apache Airflow, Databricks/Spark (Scala/Python). Vector search: Elasticsearch kNN (deployed, Ripley RAG); Pinecone, Weaviate, ChromaDB (evaluated and right-sized). Also: data labeling & RLHF (Label Studio), Tableau, Power BI.

Languages & Frameworks: Python, Java, TypeScript, JavaScript, React, Next.js, Node.js, FastAPI, Scala (limited), Uvicorn, Pydantic, SQLAlchemy, Alembic, JWT, httpx.

Infrastructure & DevOps: AWS (Nielsen platform; multi-cloud via Shadeform across AWS / GCP / Azure), GCP (Cloud Run, Cloud SQL, Cloud Logging, Secret Manager, Confidential Space, Cloud KMS), Docker / Docker Compose, Kubernetes, WebSockets, Cloudflare, Google OAuth 2.0, Fernet, resilience patterns (circuit breakers, exponential backoff, session caching), automated code-quality (pytest gates with Pydantic runtime typing, pre-commit hooks and CI verifiers where each repo runs them, linters), classification-boundary egress guards with default deny.

Agentic AI & Orchestration: Deployed multi-agent: Ripley (LangChain routing + SuperAGI autonomy, a Compound AI System), EGON and Project E (Google ADK + Google's A2A protocol), Lynda (36-tool governed assistant plus Satchel). Agent harnesses: coding agent under its own identity with owner-merged PRs, cross-family judging of plans and diffs, validator-gated research agents, after-answer truth checks. Graph-grounded output verification. Single-agent: Pydantic-driven structured tool use, bounded agent loops, Human-in-the-Loop (HITL) workflows, MCP. Worked with: LangGraph, CrewAI, AutoGen, Semantic Kernel, LlamaIndex.

ML Ops & AI Systems: LLM observability & evaluation (Arize, Weights & Biases, promptfoo), AWS & Google ML Ops platforms, model evaluation, capability auditing, AI failure-mode analysis, boundary testing, cross-family attack drills.

AI-Augmented Engineering: Claude Code (2025–present), Wispr Flow, VS Code, context-aware coding, AI pair programming, prompt engineering, AI workflow automation, prototyping / MVP.

Voice & Multimodal: Conversational AI / VUI, ASR/TTS, Vapi, Retell AI, streaming audio pipelines, WebSpeech, VAD, Chirp, Groq, ElevenLabs, Deepgram.

Data Strategy & Governance: NIST AI RMF, NIST CSF 2.0, SR 11-7 (model risk management), SEC/FINRA & NAIC alignment, FIBO and DoD program-element / PSC ontologies, data provenance & lineage, PII governance & RBAC, Explainable AI (XAI), audit-trail architectures, data sovereignty, confidential computing (TEEs), data contracts, data lifecycle management, secure data pipelines, AI ethics & safety policy.